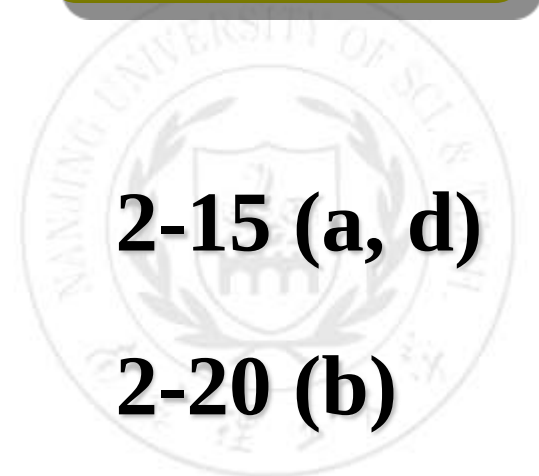
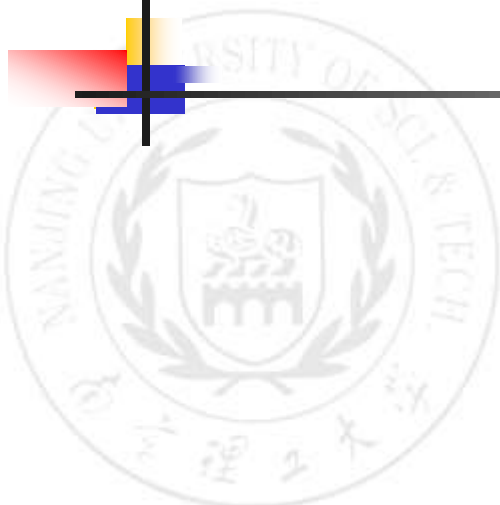


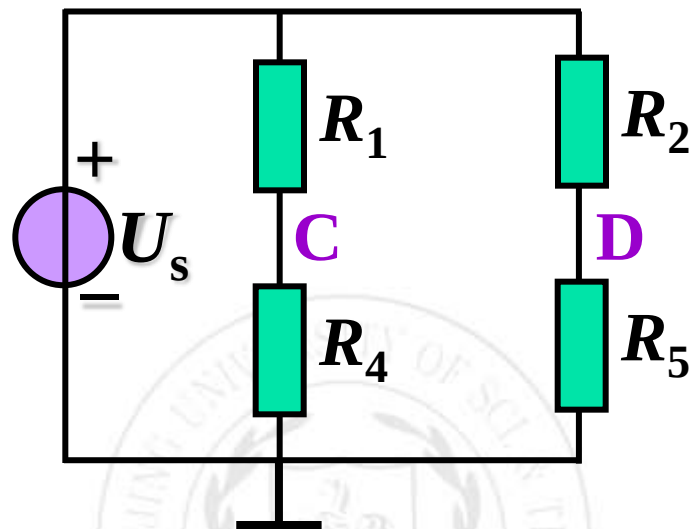


作业

2-15 (a, d)

2-20 (b)

2-21 (a)

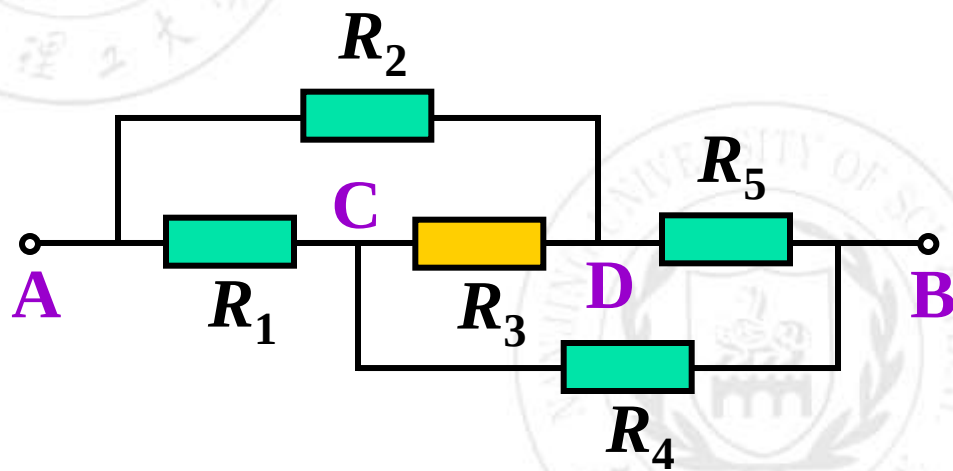
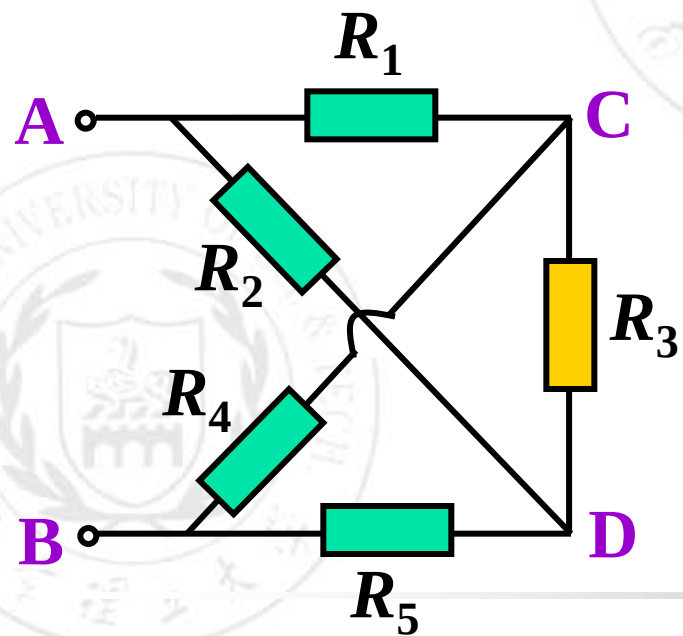
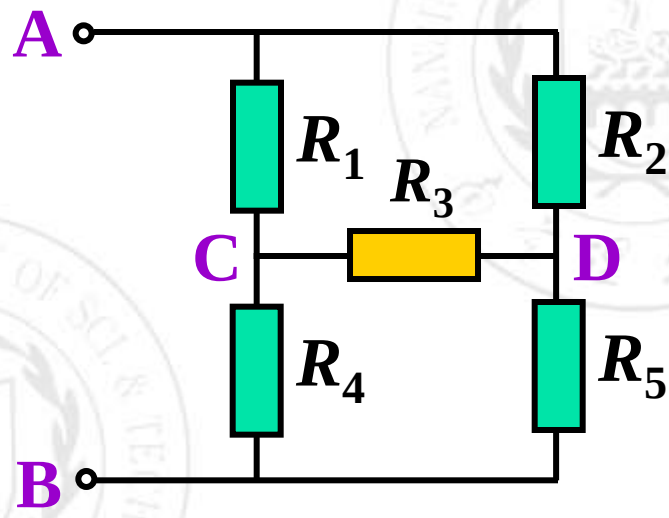
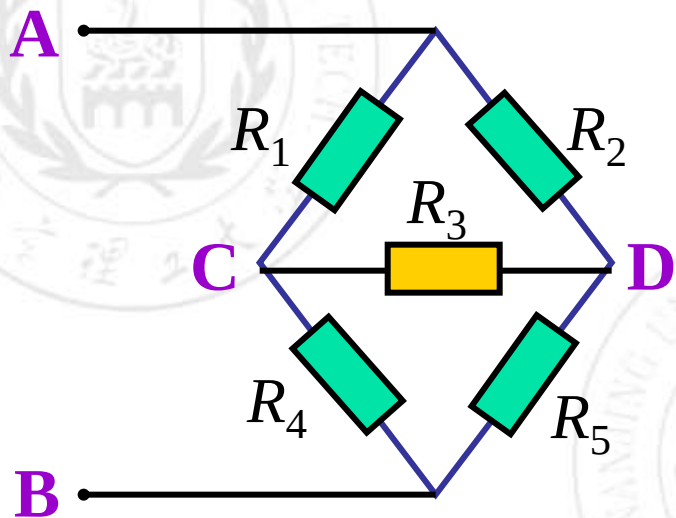


$$U_C = \frac{R_4}{R_1 + R_4} U_s, U_D = \frac{R_5}{R_2 + R_5} U_s$$

当 $R_1 R_5 = R_2 R_4$ $\Rightarrow$ $U_C = U_D$ 等电位点

在等电位点之间接入任何电阻都不会对其他支路产生影响。

电桥电路

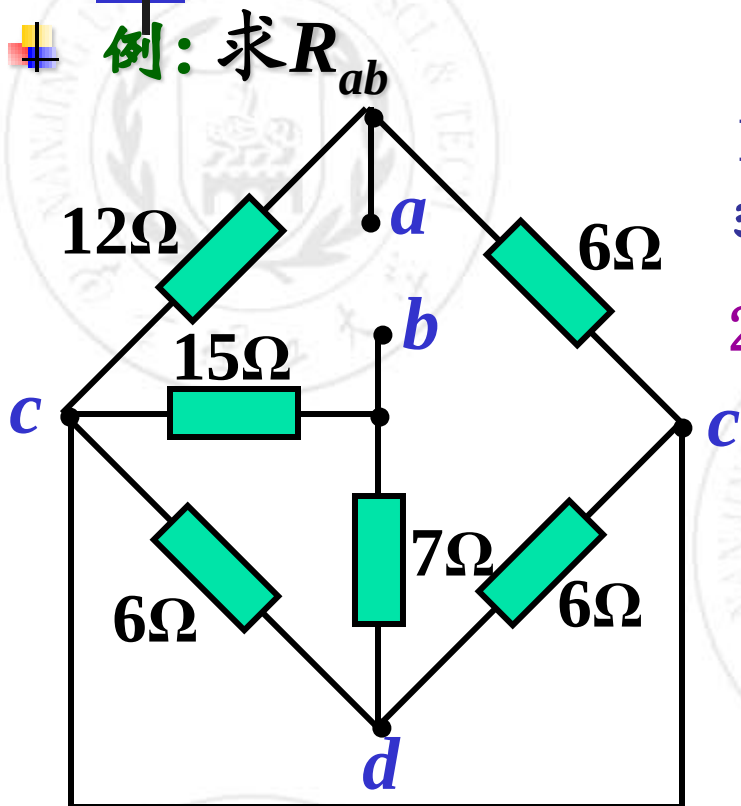


2.2 电阻的串联、并联和混联

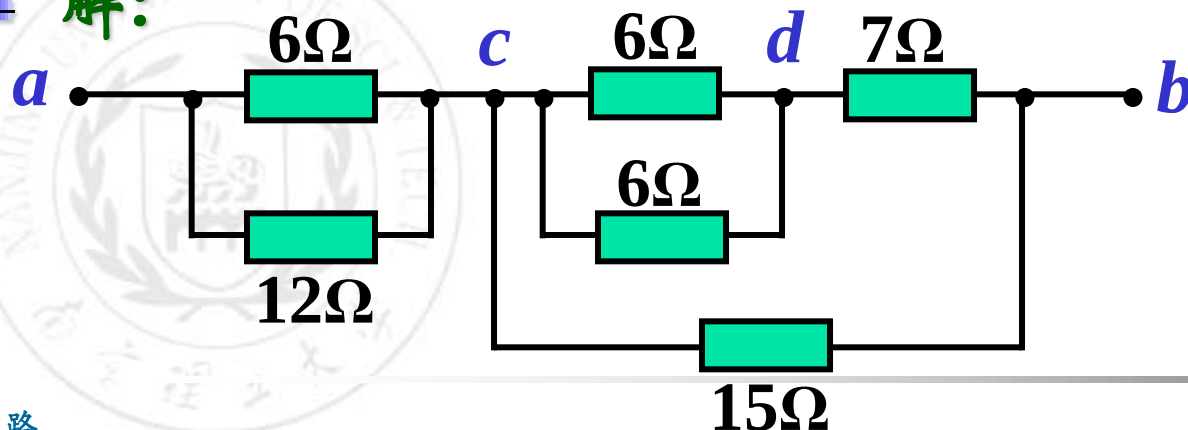
字母标注法

1、在各节点处标上节点字母，短路线联接的点或等位点用同一字母标注；

2、整理并简化电路，求出总的等效电阻。



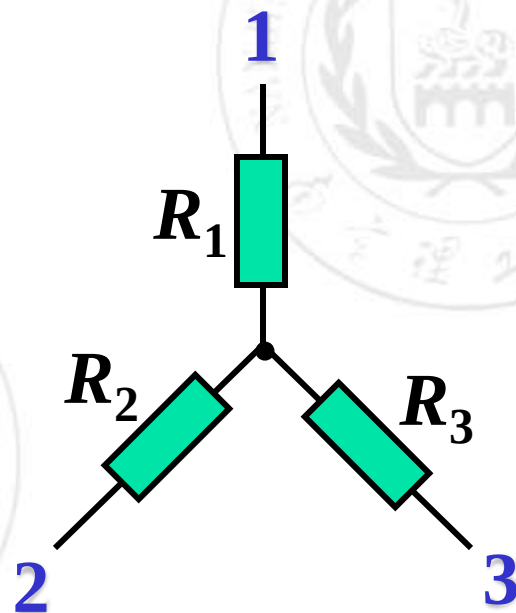
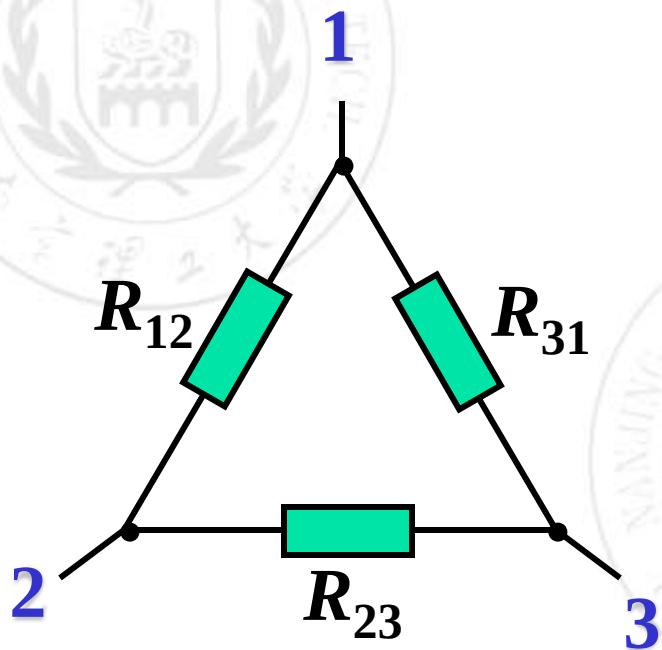
解：



$$R_{ab} = 4 + 6 = 10\Omega$$

2.3 电阻的Y-△等效变换

■ 等效互换的公式



$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3};$$

$$R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}}$$

当三个电阻相等时, $R_{\Delta} = 3R_Y$ 。

目 录

2.1 等效变换的概念

2.2 电阻的串联、并联和混联

2.3 电阻的Y- Δ 等效变换

2.4 电压源、电流源的串联和并联

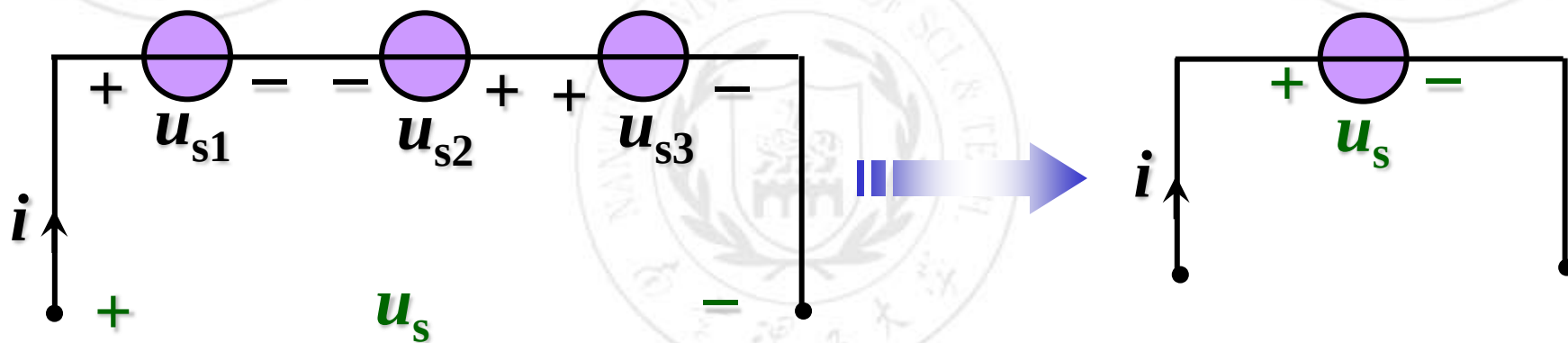
2.5 实际电源的等效变换

2.6 运用等效变换分析含受控源的电阻电路



2.4 电压源、电流源的串联和并联

■ 电压源的串联

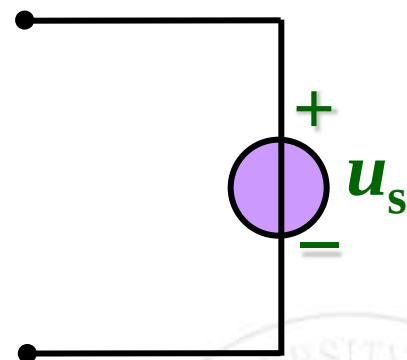
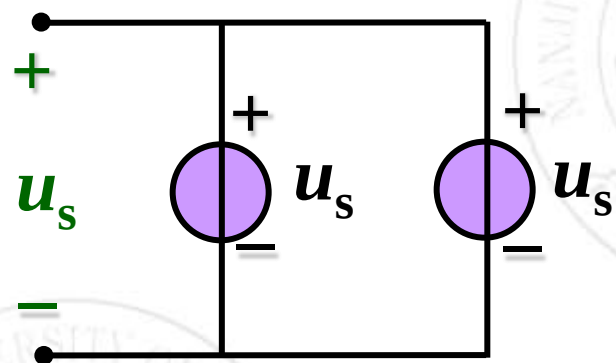


$$u_s = u_{s1} - u_{s2} + u_{s3} = \sum u_{sk}$$

2.4 电压源、电流源的串联和并联

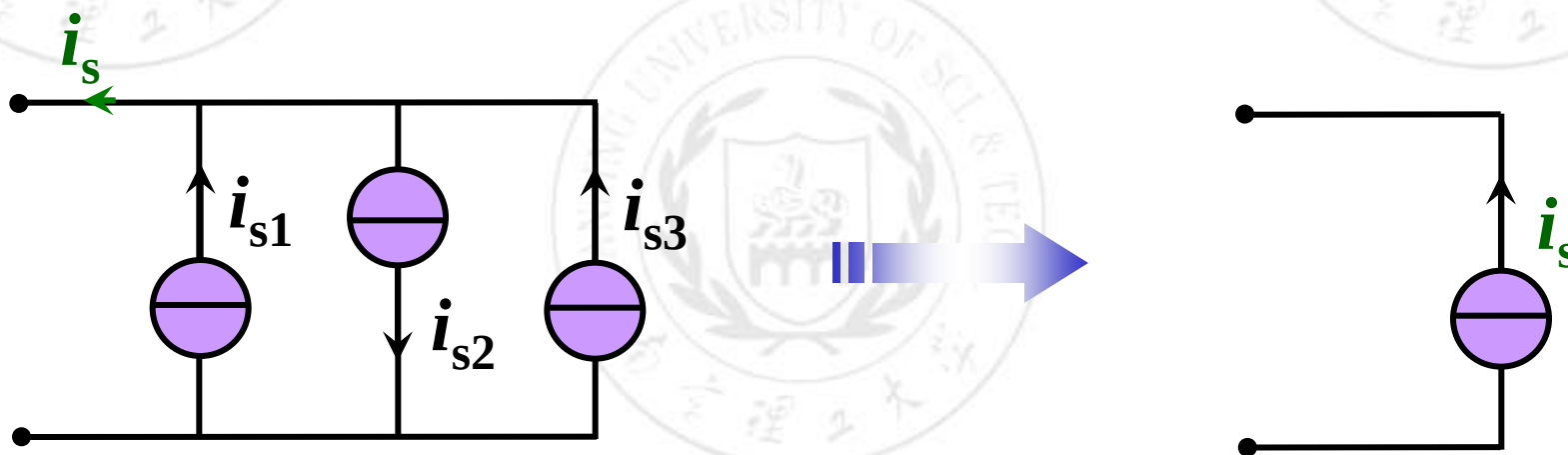
■ 电压源的并联

■ 同极性、同数值并联



2.4 电压源、电流源的串联和并联

■ 电流源的并联

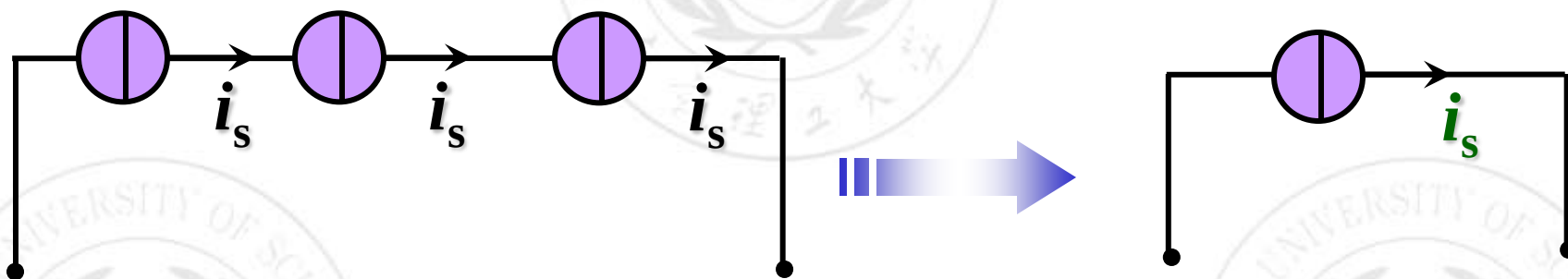


$$i_s = i_{s1} - i_{s2} + i_{s3} = \sum i_{sk}$$

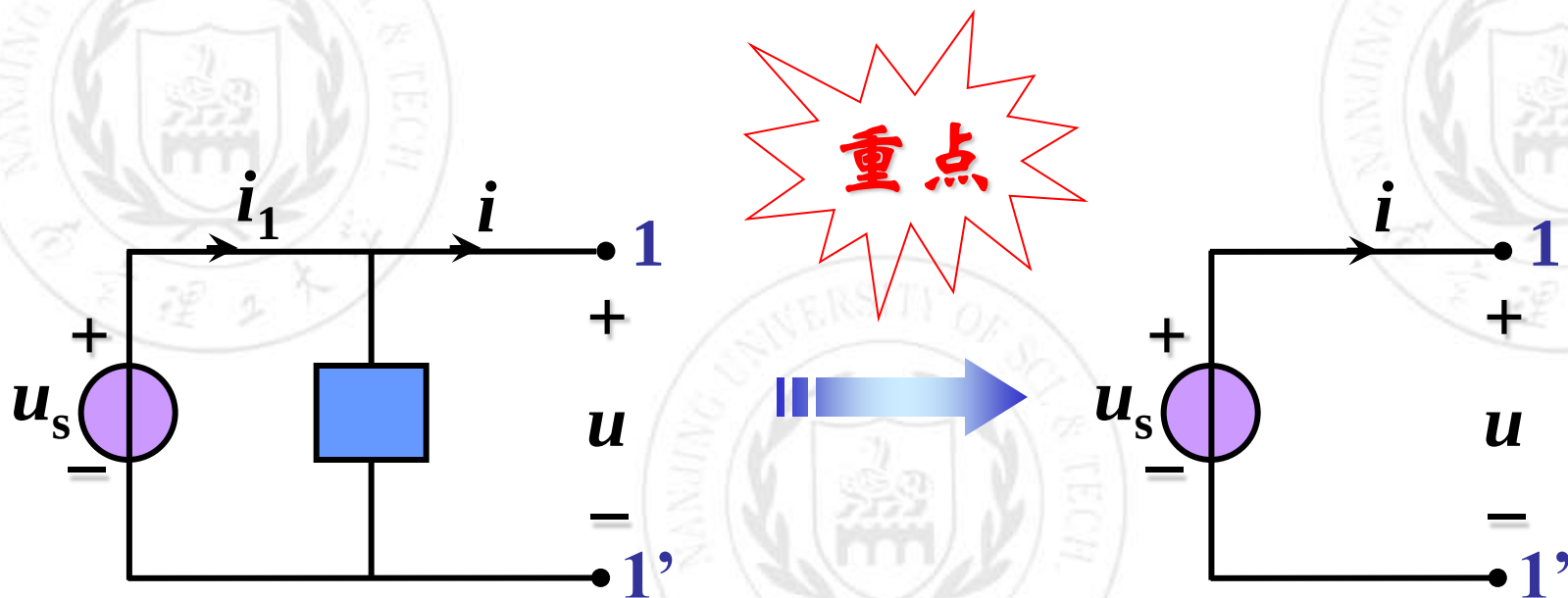
2.4 电压源、电流源的串联和并联

■ 电流源的串联

■ 同方向、同数值串联



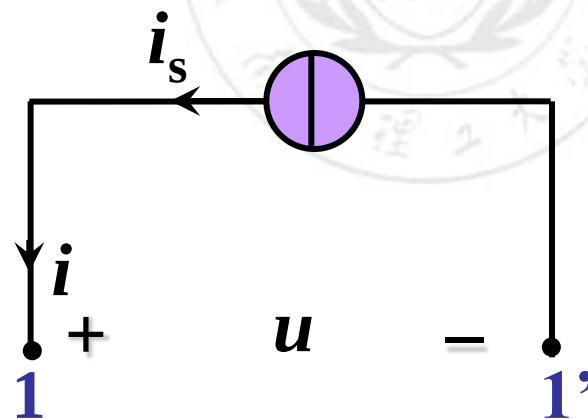
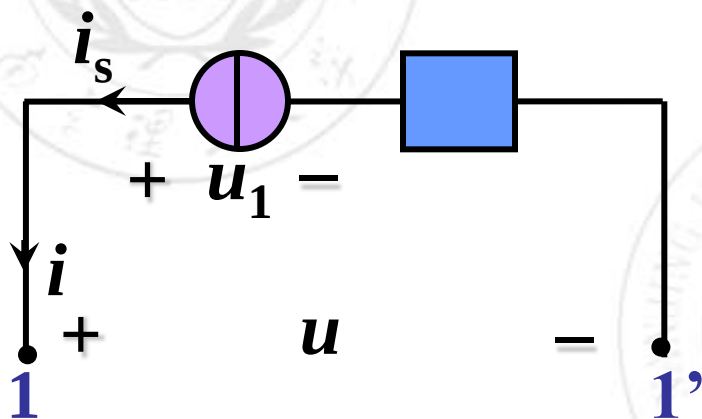
2.4 电压源、电流源的串联和并联



与理想电压源**并联**的元件（支路）对外电路讨论时可**断开**。

2.4 电压源、电流源的串联和并联

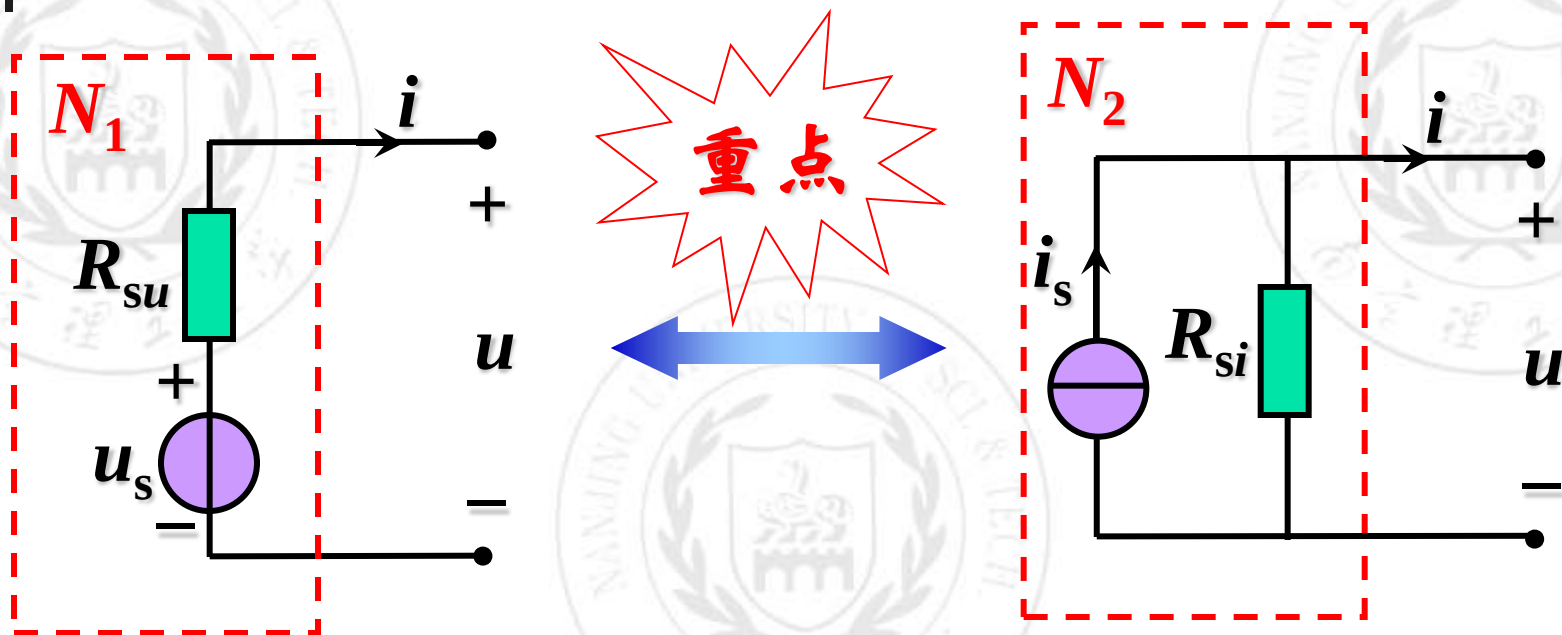
重点



与理想电流源 **串联** 的元件（支路）对外电路讨论时可 **短接**。



2.5 实际电源的等效变换



N_1 : $u = u_s - R_{su} i$

N_2 : $i = i_s - \frac{u}{R_{si}}$
 $u = R_{si} i_s - R_{si} i$

$$\Rightarrow \begin{cases} R_{su} = R_{si} \triangleq R_s \\ u_s = R_s i_s \quad (i_s = \frac{u_s}{R_s}) \end{cases}$$

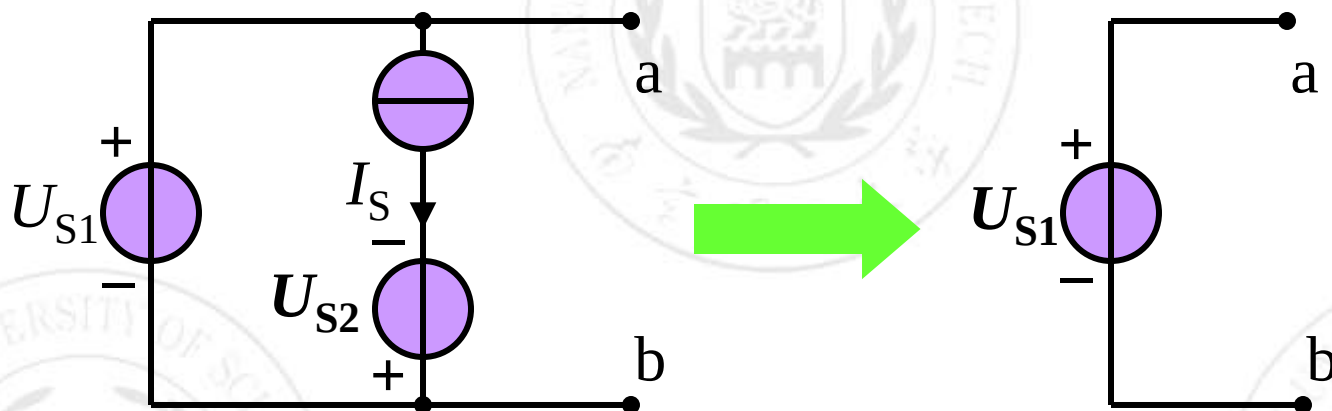
■ 注意！

- 保持变换前后参考方向**一致**.
- 等效是对外部而言，对内不等效.
- 理想电压源和理想电流源之间没有等效关系.

2.5 实际电源的等效变换

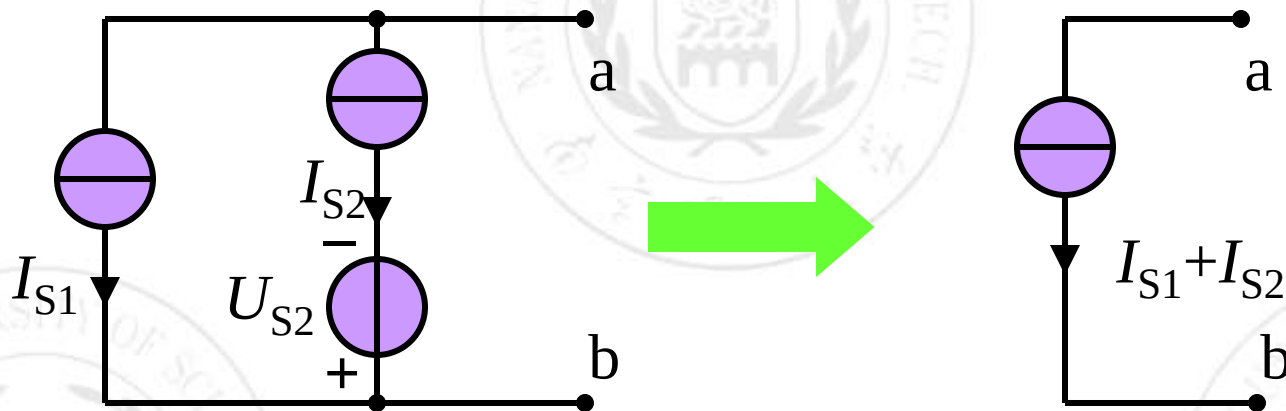
例、将下列电路简化成最简单的电路：

1



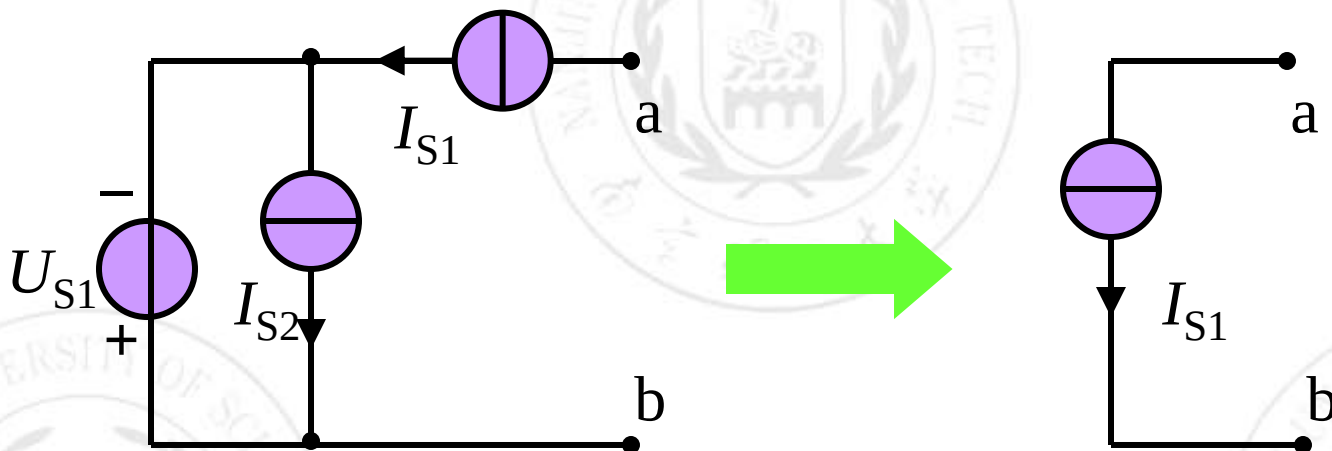
例、将下列电路简化成最简单的电路：

2



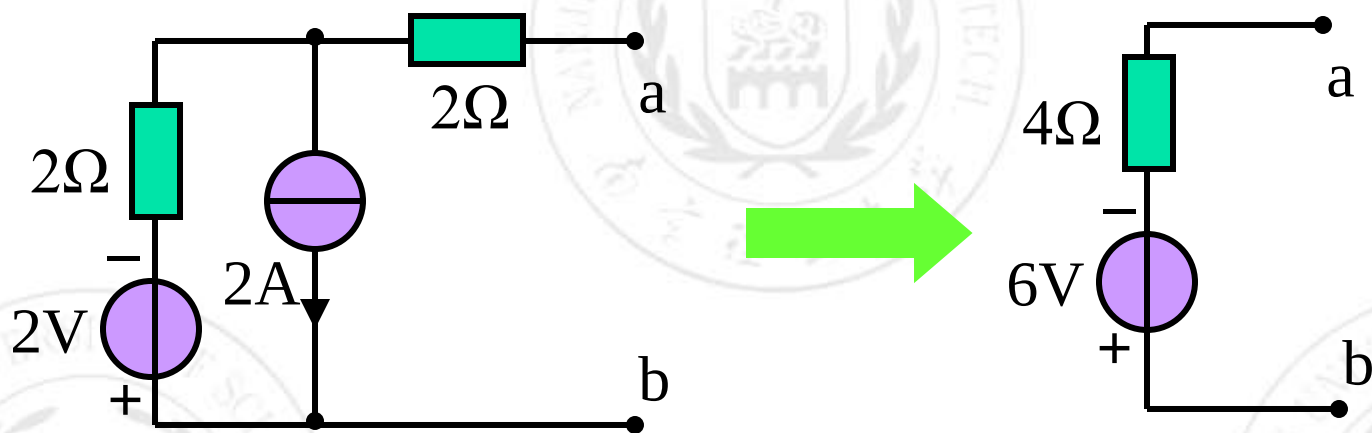
例、将下列电路简化成最简单的电路：

3



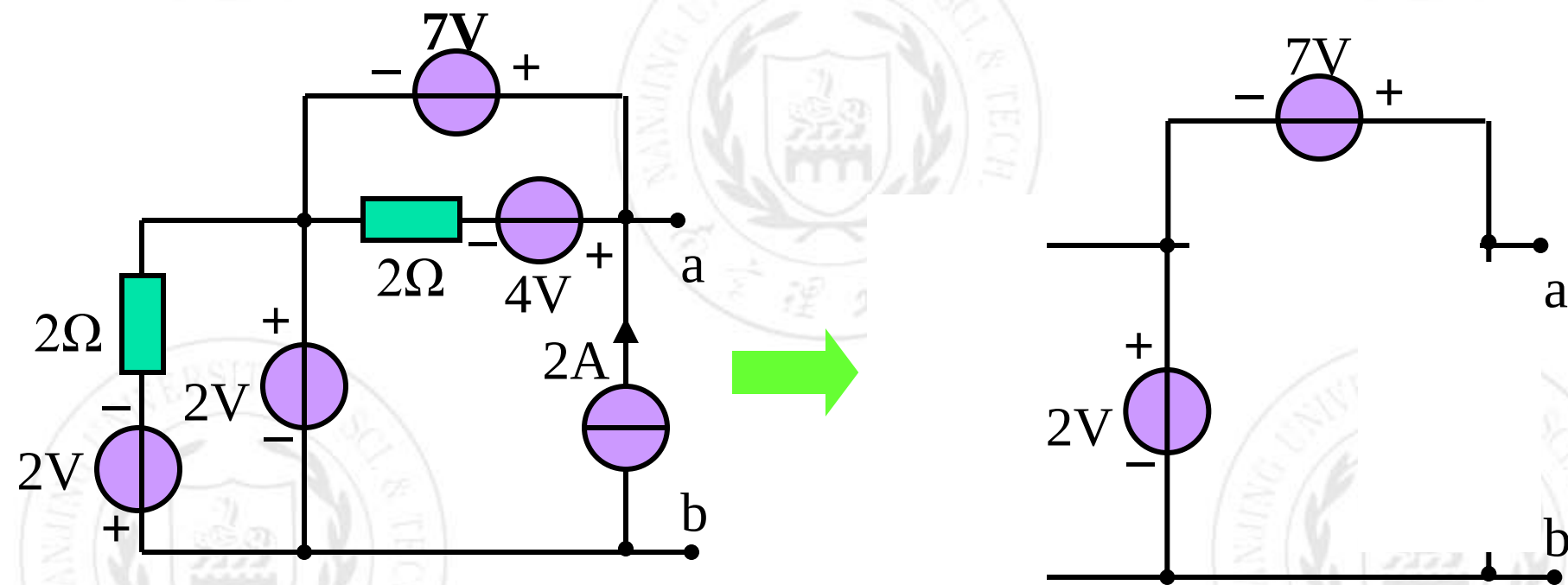
例、将下列电路简化成最简单的电路：

4



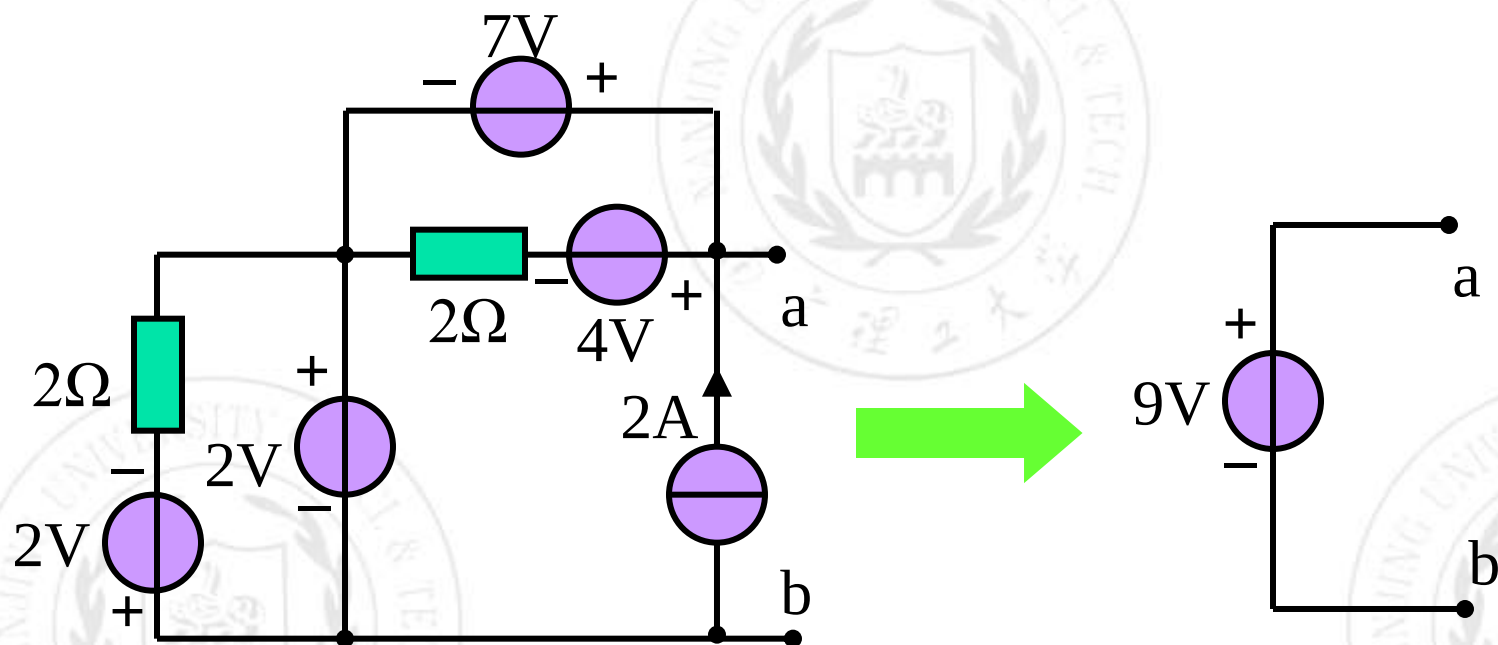
2.5 实际电源的等效变换

例、将下列电路简化成最简单的电路：



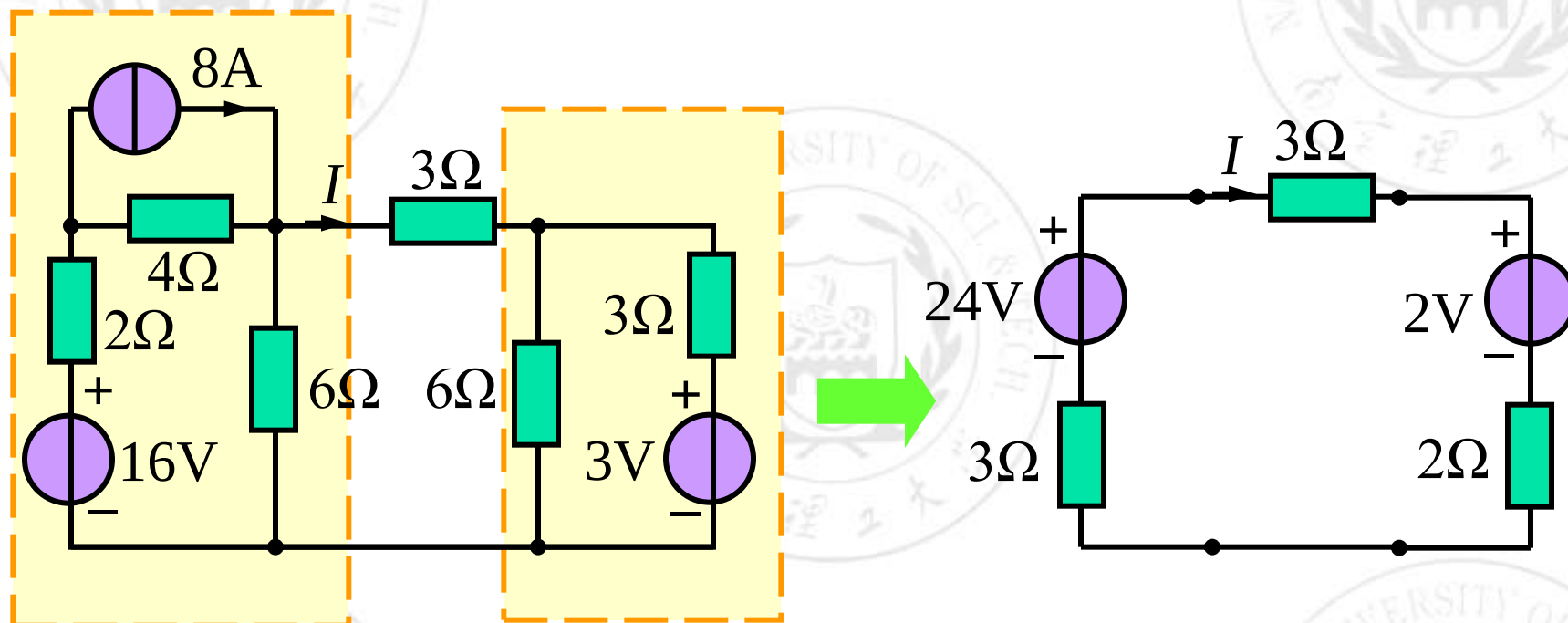
2.5 实际电源的等效变换

例、将下列电路简化成最简单的电路：



2.5 实际电源的等效变换

例、求电流 I 。

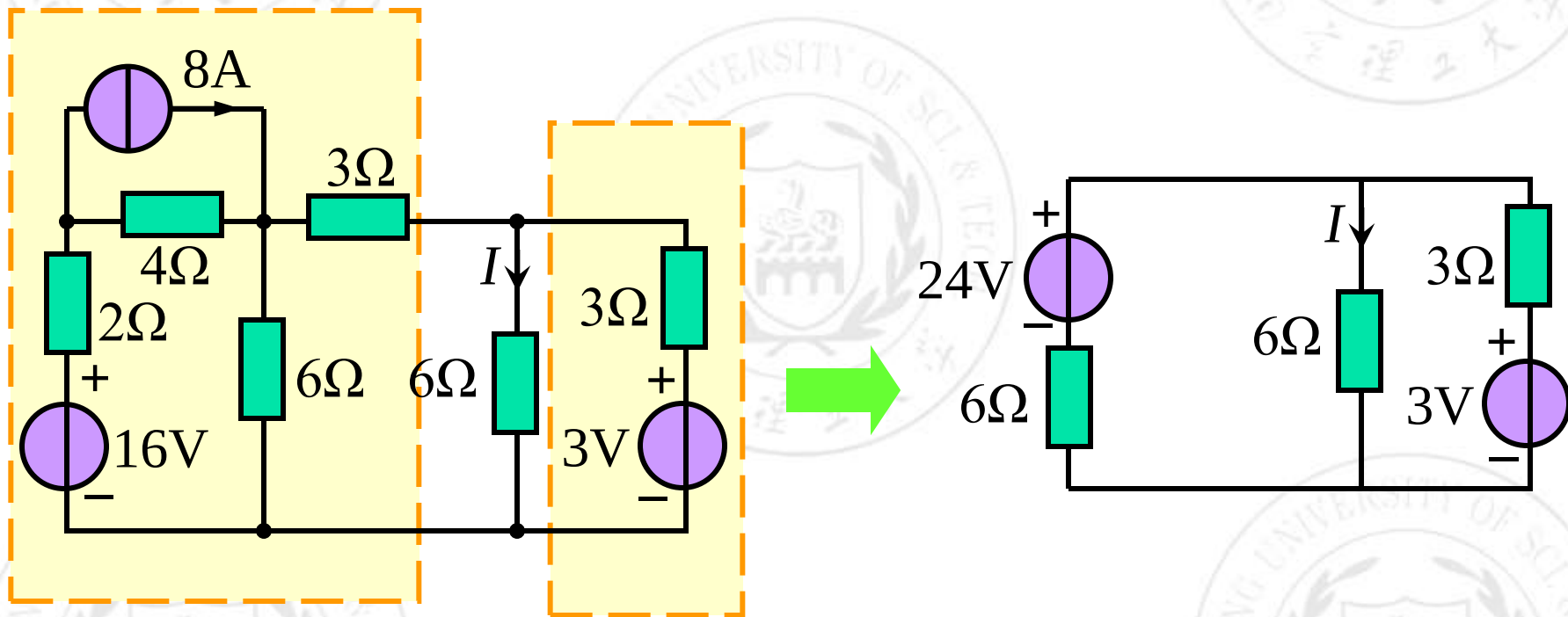


得：

$$I = \frac{24 - 2}{3 + 3 + 2} = 2.75 \text{ A}$$

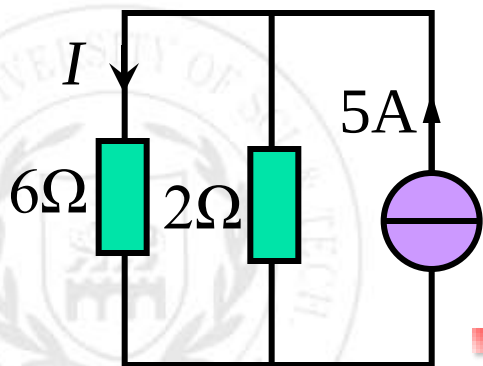
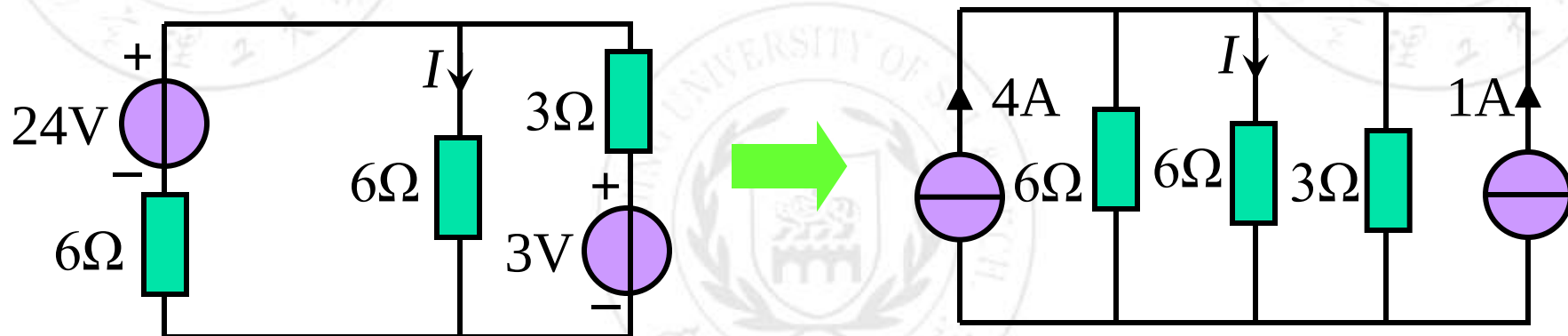
2.5 实际电源的等效变换

例、求电流 I 。



2.5 实际电源的等效变换

例、电流 I 。



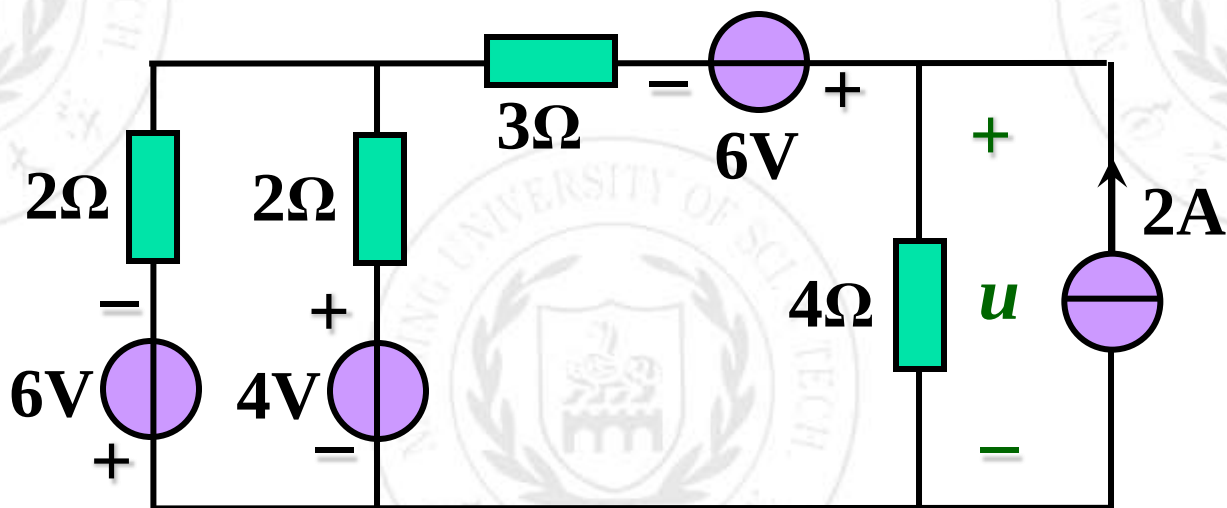
$$I = \frac{2}{2+6} \times 5 = 1.25\text{A}$$



注意：未知量所在的支路一般保持不动

2.5 实际电源的等效变换

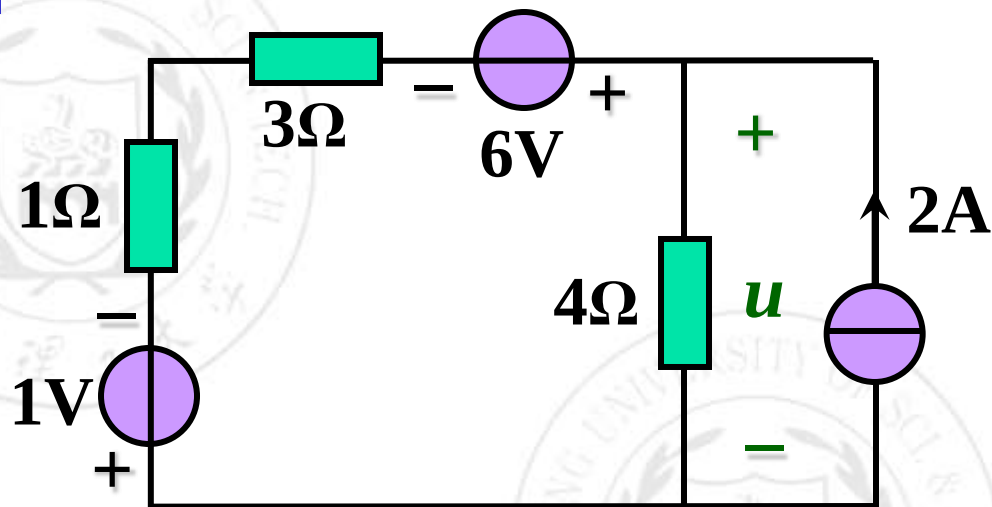
例：运用电源等效变换方法求 u



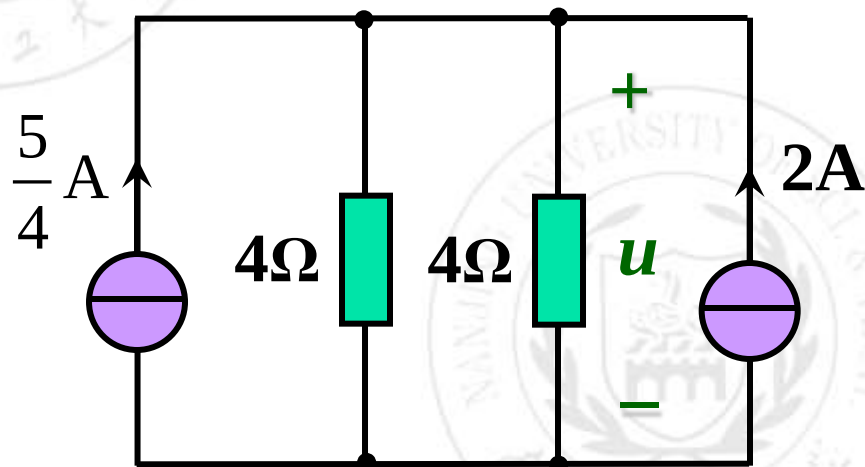
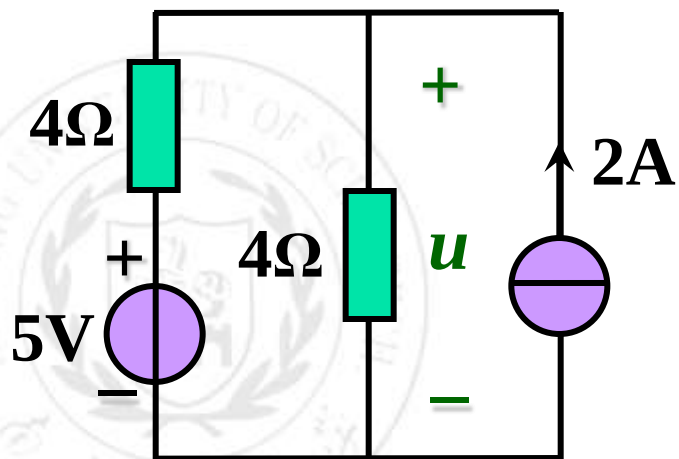
解：

$$u = 4 \times \frac{2 + \frac{5}{4}}{2} = 6.5V$$

2.5 实际电源的等效变换

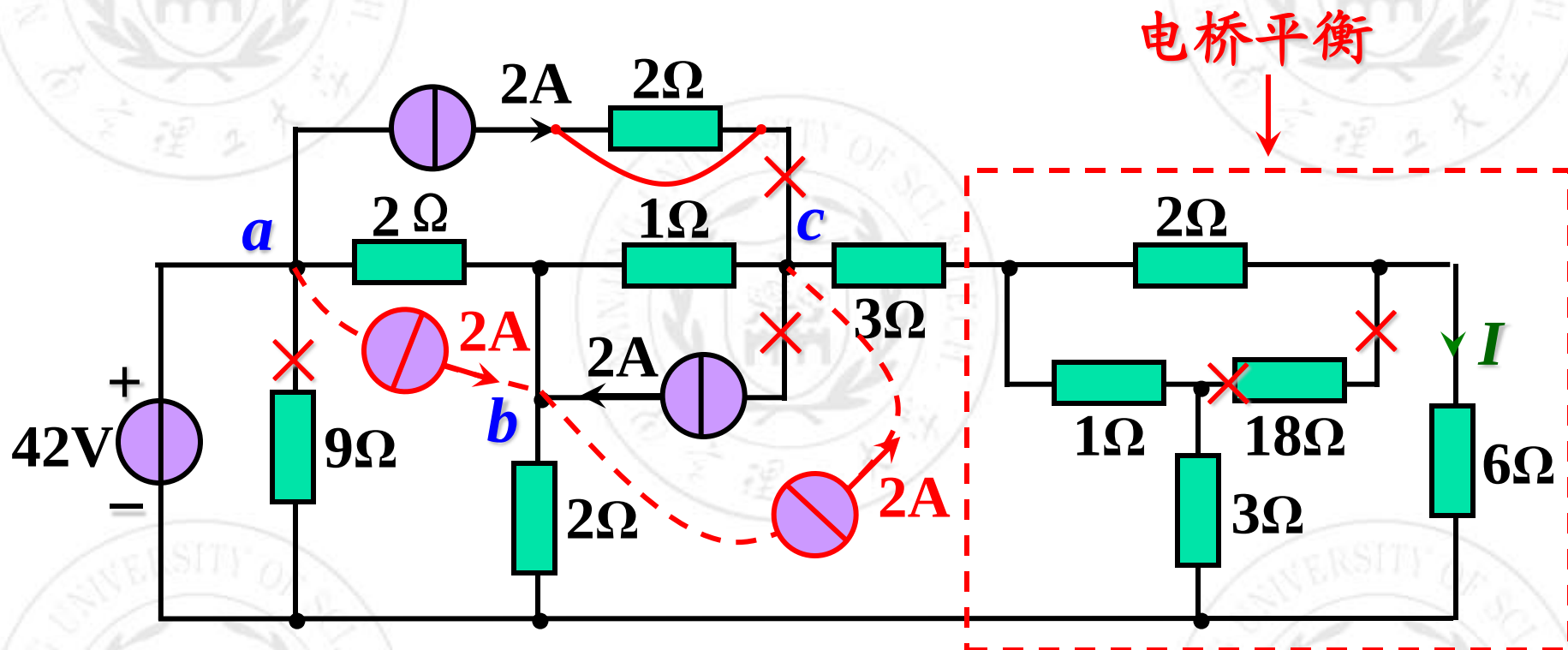


$$u = 4 \times \frac{2 + \frac{5}{4}}{2} = 6.5\text{V}$$

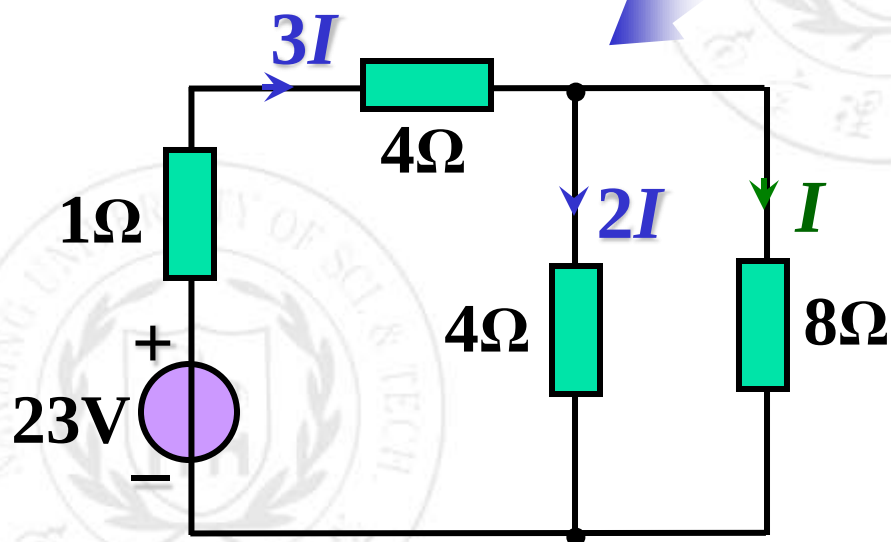
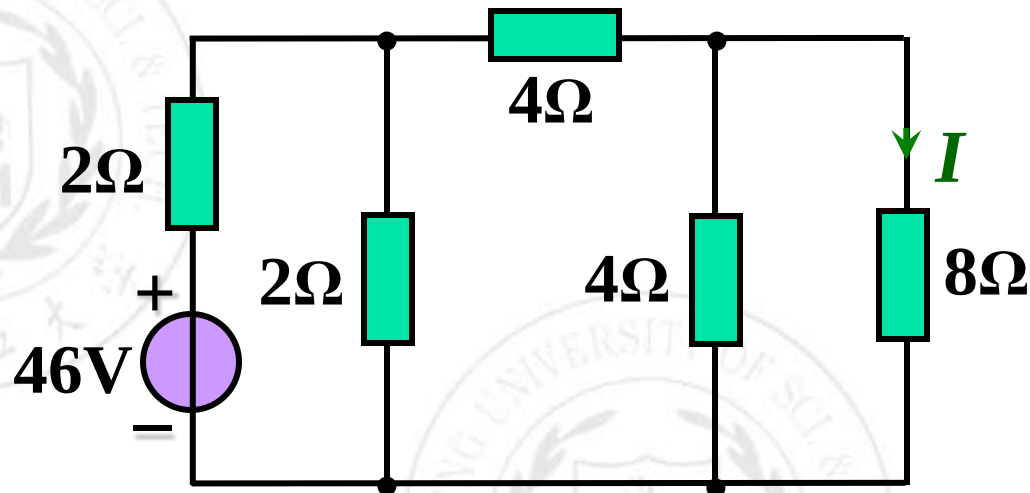


2.5 实际电源的等效变换

例：运用等效变换方法求 I



2.5 实际电源的等效变换



$$5 \times 3I + 8I = 23 \Rightarrow I = 1A$$



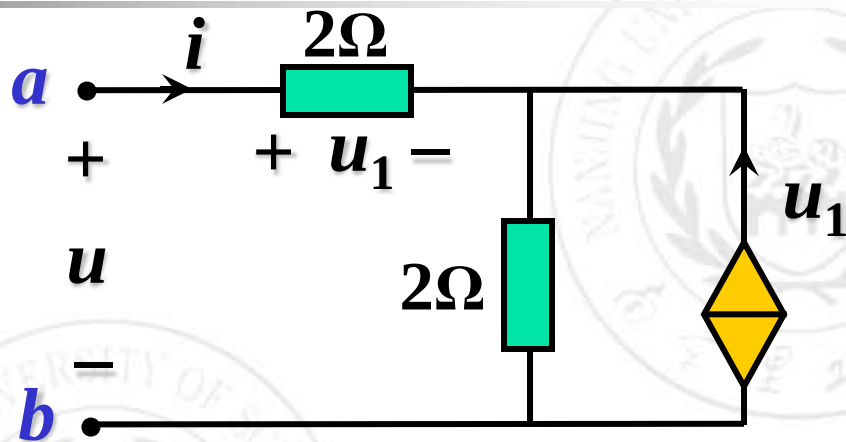
■ 用等效变换方法分析含受控源电路

- 求含受控源一端口电阻电路的输入电阻时，一律用欧姆定律。
- 受控源与电阻串并联等效变换与独立源类似。
- **注意：**等效变换中**控制支路**尽量不参与变换。

2.6 运用等效变换分析含受控源的电阻电路

例：求输入电阻

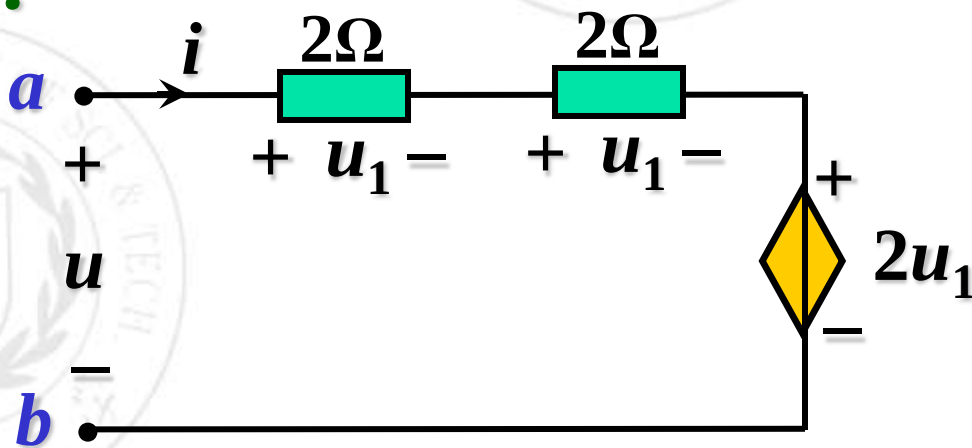
重点



解：输入电阻：

$$R_{in} = \frac{u}{i} = \frac{u_1 + 2 \times 1.5u_1}{\frac{u_1}{2}} = 8\Omega$$

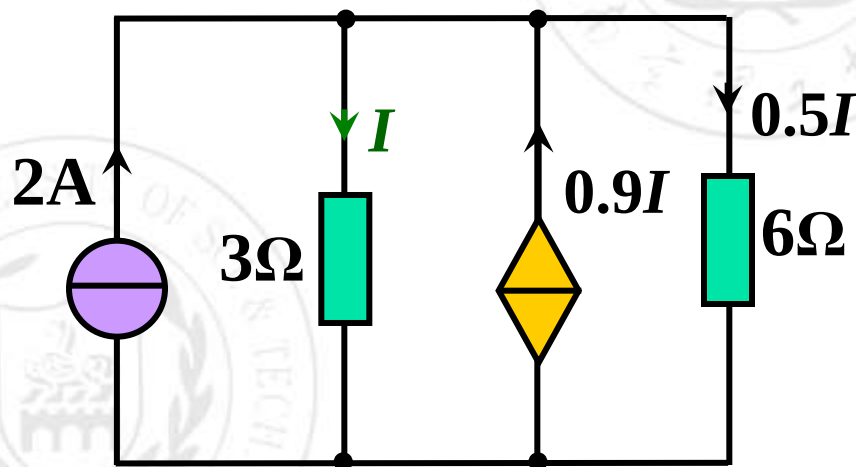
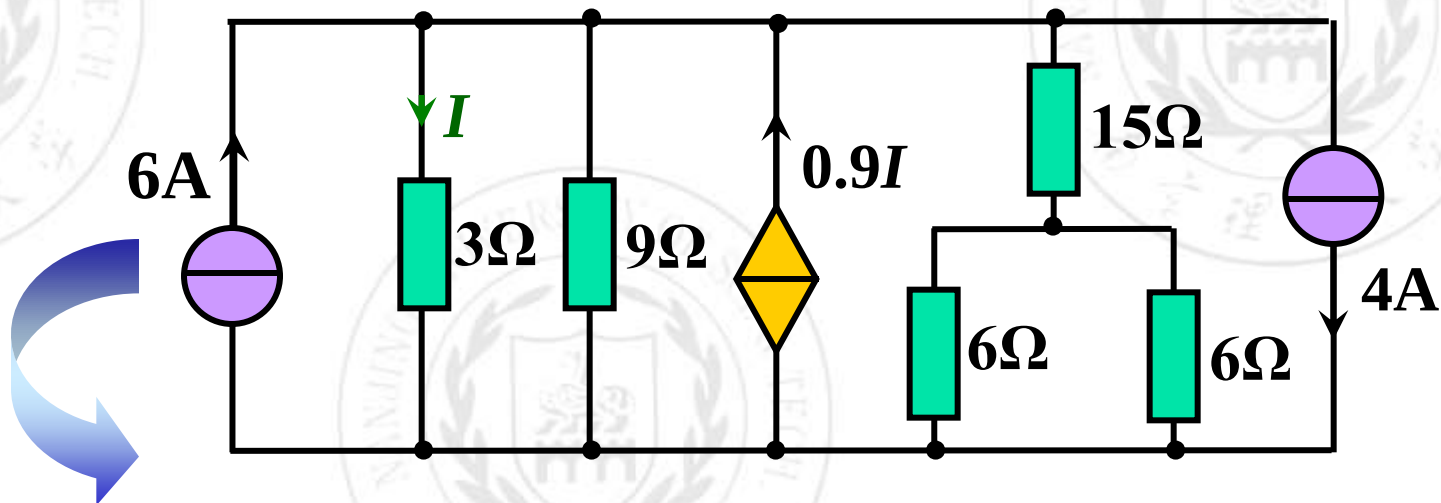
另解：



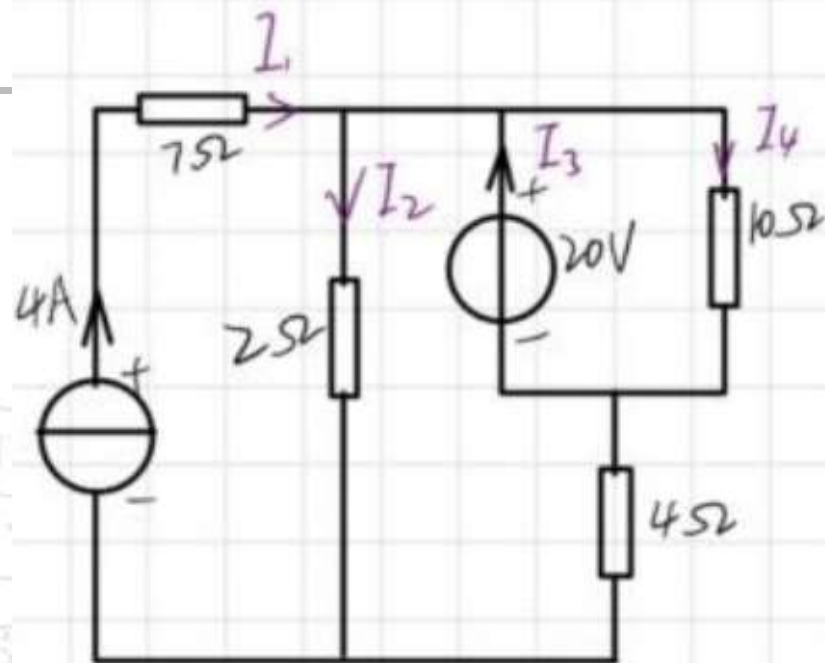
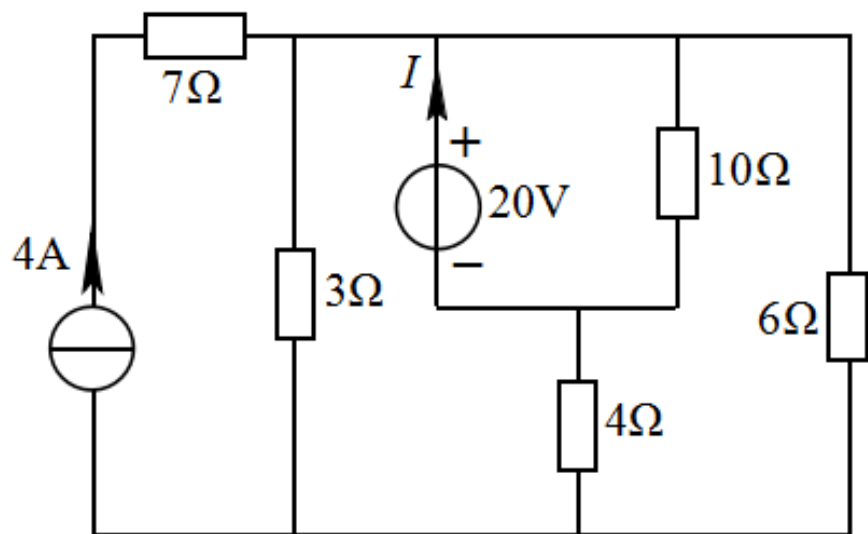
$$R_{in} = \frac{4u_1}{0.5u_1} = 8\Omega$$

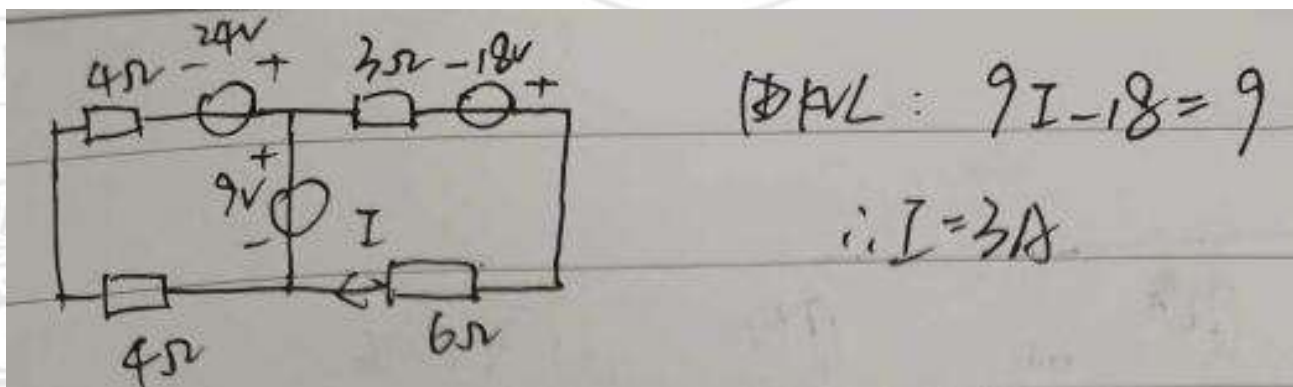
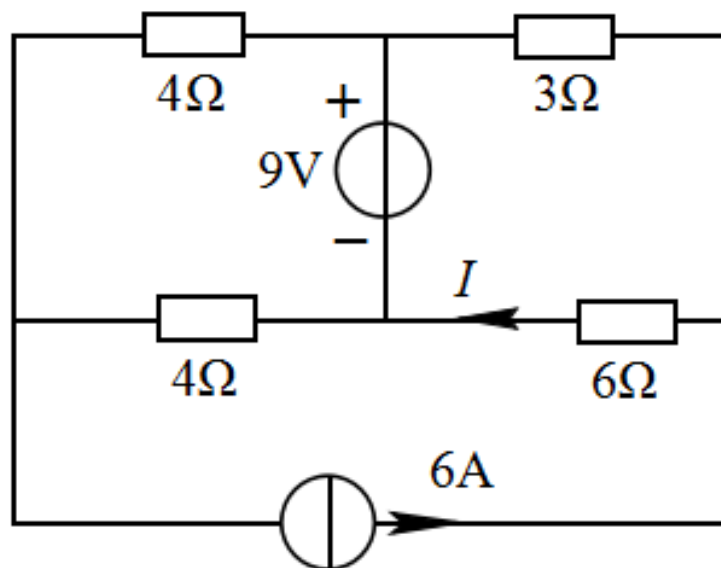
2.6 运用等效变换分析含受控源的电阻电路

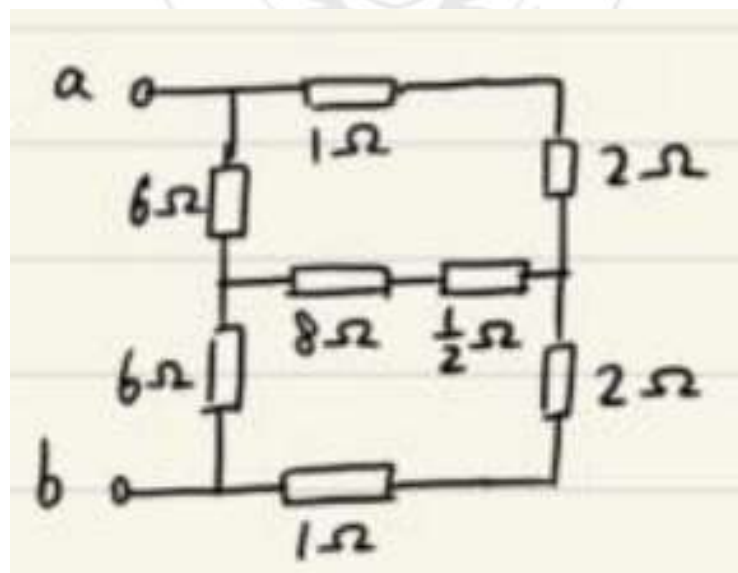
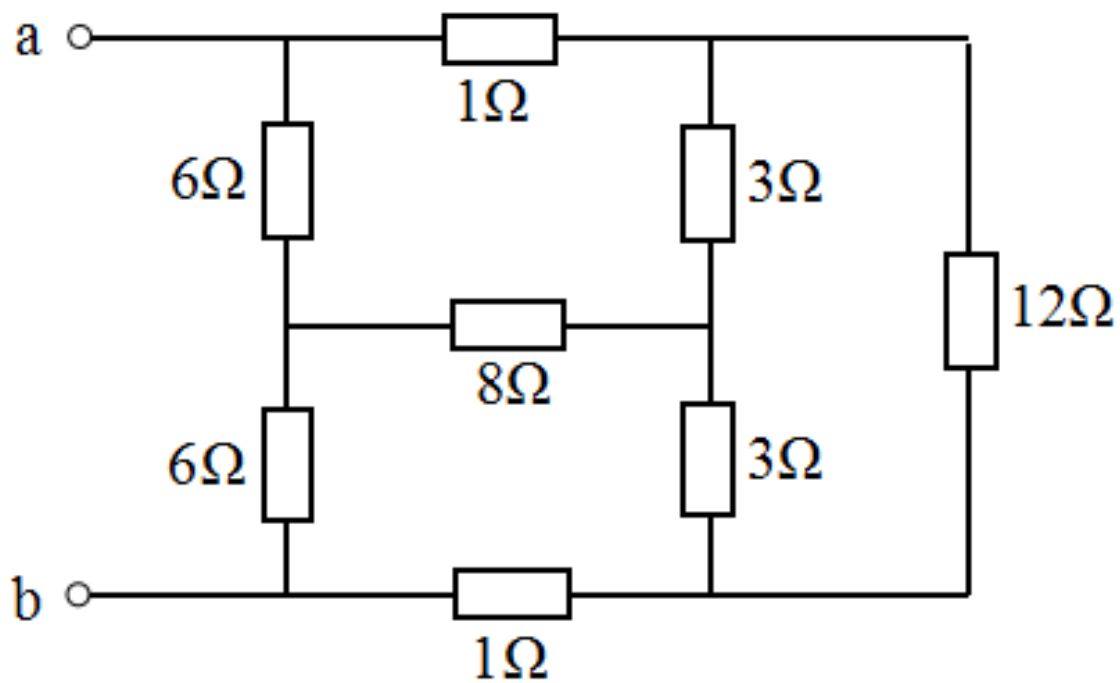
例：运用等效变换方法I



$$I + 0.5I - 0.9I = 2$$
$$I = \frac{10}{3} \text{ A}$$







目 录

3.1 支路电流法 

3.2 网孔电流法和回路电流法

3.3 节点电压法

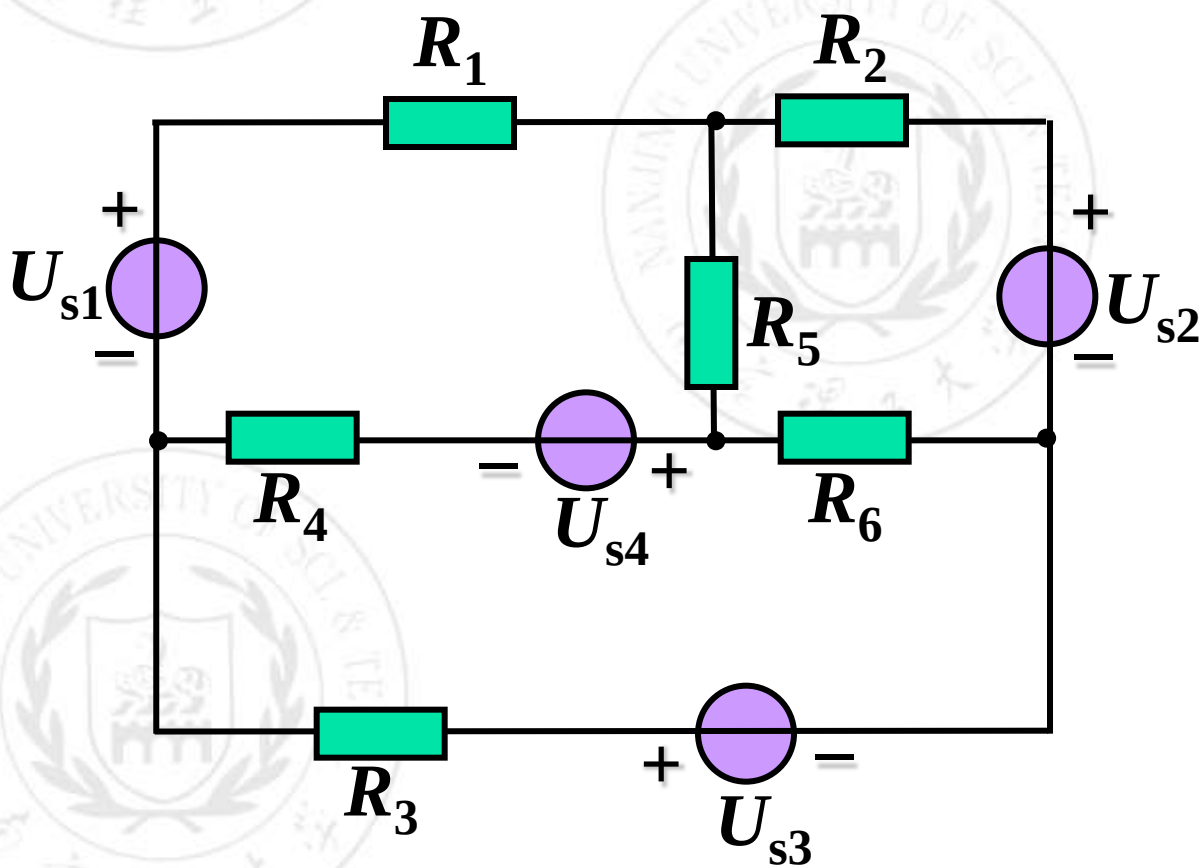
■ 支路电流法

✚ 以支路电流为未知量，根据KCL、KVL列关于支路电流的方程，进行求解的过程。

3.1 支路电流法

基本步骤

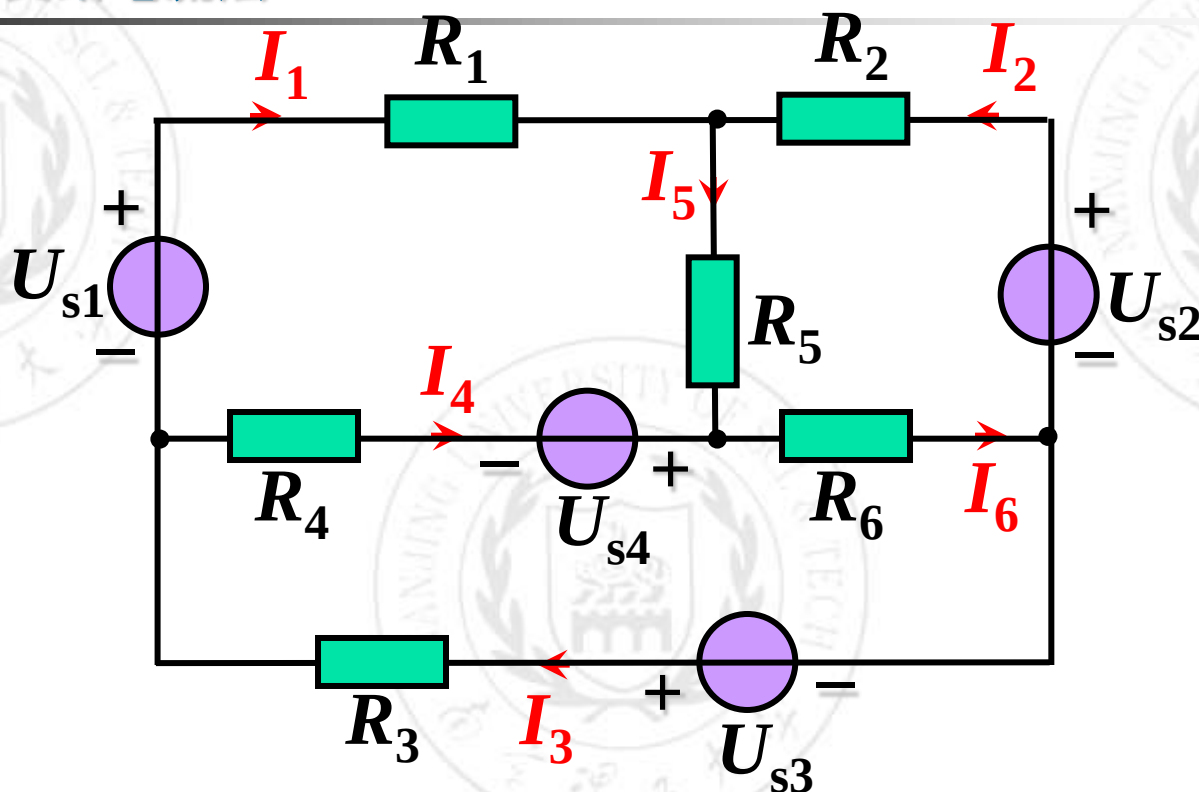
■ 仅含电阻和电压源的电路



节点数(n): 4

支路数(b): 6

3.1 支路电流法



■ **第1步:** 选定各支路电流参考方向, 各节点KCL方程如下:

节点 1: $I_1 - I_3 + I_4 = 0$

节点 2: $-I_1 - I_2 + I_5 = 0$

节点 3: $I_2 + I_3 - I_6 = 0$

节点 4: $-I_4 - I_5 + I_6 = 0$

3.1 支路电流法

$$1: I_1 - I_3 + I_4 = 0$$

$$2: -I_1 - I_2 + I_5 = 0$$

$$3: I_2 + I_3 - I_6 = 0$$

$$4: -I_4 - I_5 + I_6 = 0$$

可见：上述四个节点的KCL方程不是相互独立的

若选图中所示电路中的节点4为参考节点，则节点1、2、3为独立节点，其对应的KCL方程必将独立，

即：

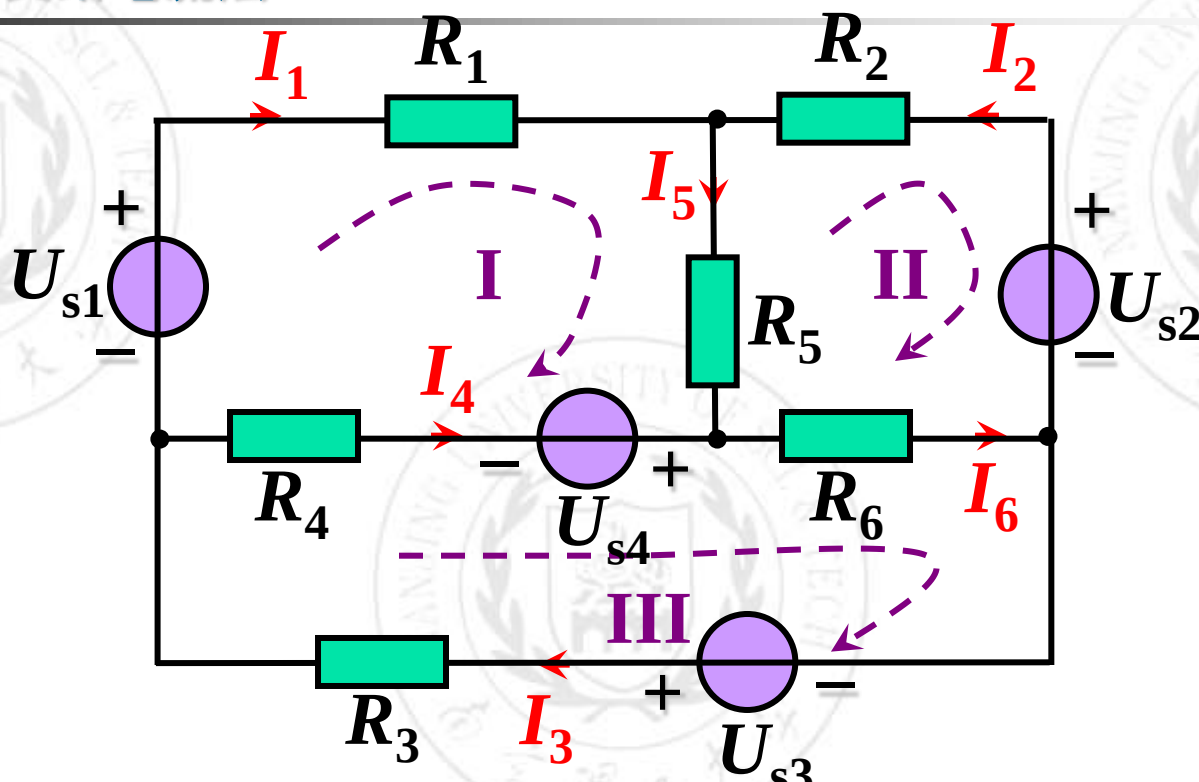
$$1: I_1 - I_3 + I_4 = 0$$

$$2: -I_1 - I_2 + I_5 = 0$$

$$3: I_2 + I_3 - I_6 = 0$$

第2步：对 $(n-1)$ 个独立节点列KCL方程

3.1 支路电流法



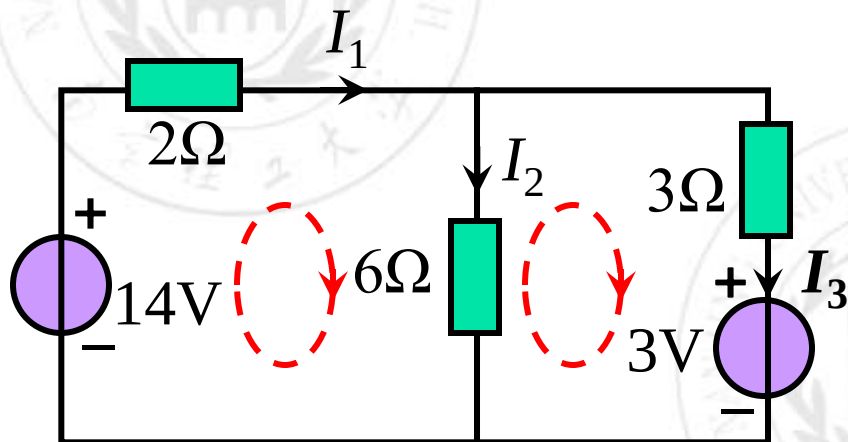
■ 第3步：对 $b-(n-1)$ 个独立回路列关于支路电流的KVL方程

$$\text{I: } R_1 I_1 + R_5 I_5 + U_{s4} - R_4 I_4 - U_{s1} = 0$$

✚ 演算：找没有列过KCL方程的节点和没有列过KVL方程的回路验证。

3.1 支路电流法

例：用支路电流法求电路中各支路电流。



**解：首先设定支路电流，
列写 $n-1$ 个KCL方程。**

$$I_1 = I_2 + I_3$$

其次列写 $b-(n-1)$ 个KVL方程。

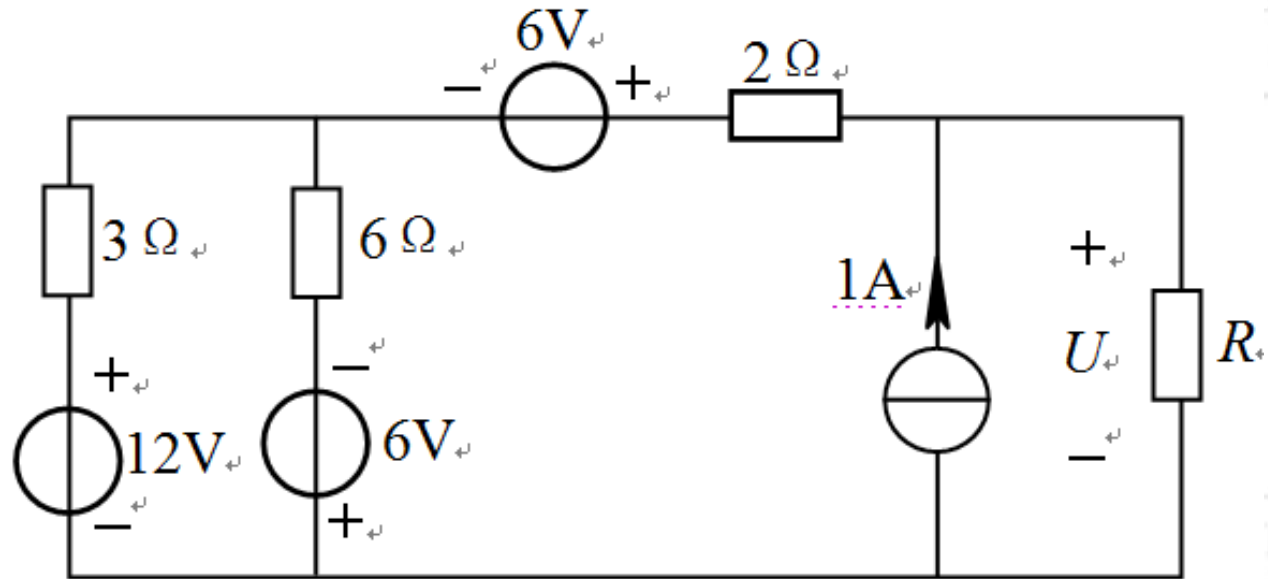
$$2I_1 + 6I_2 - 14 = 0$$

$$3I_3 + 3 - 6I_2 = 0$$

最后联立方程解得：

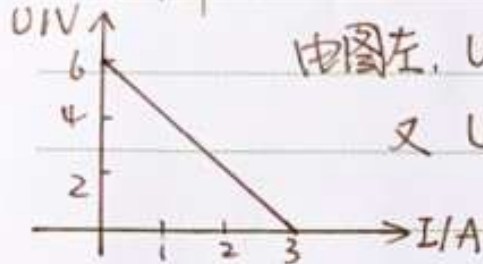
$$I_1 = 3\text{A} \quad I_2 = \frac{4}{3}\text{A} \quad I_3 = \frac{5}{3}\text{A}$$

电路如图所示, 试用电源等效变换的方法求 $U=8V$ 时, $R=?$



$$R = 4\Omega$$

2-1b. 解:



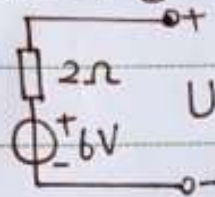
由图左, $U = -2I + 6$ 或 $I = -\frac{U}{2} + 3$.

又 $U = U_s - RI$, $I = I_s - GU$.

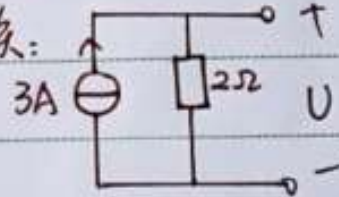
得 $U_s = 6V$, $R = 2\Omega$

$I_s = 3A$, $G = \frac{1}{2}S \Rightarrow R = 2\Omega$.

$\therefore N_s$ 最简单的模型为: 串联.



并联:



课堂练习

- ◆ 2-14 (d)
- ◆ 2-19(a, c)



本次课重点

- ◆ 电源的等效变换.
- ◆ 受控源电路的等效变换.
- ◆ 支路电流法.